Distribution Management System

Enterprise-Grade Distribution & Supply Chain Platform

Platform Desktop | Mobile | Web Version 1.0.0 License Commercial

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1. Executive Summary

The **Distribution Management System (DMS)** is a comprehensive, enterprise-grade software platform purpose-built for wholesale distributors, FMCG companies, pharmaceutical distributors, and any business managing multi-route, multi-salesman field distribution operations.

The platform unifies **three critical business layers** into a single ecosystem:

Layer	Platform	Purpose

Layer	Platform	Purpose
Command Center	Windows Desktop Application	Full operational control — inventory, orders, finances, analytics
Field Operations	Android Mobile Application	Offline-first order-taking for salesmen in the field
Cloud Backbone	Centralized API Server	Real-time data sync, business logic, persistent storage

At its core, the DMS replaces 5–7 separate software tools — order management, warehouse management, invoicing, delivery tracking, field sales, ledger/accounting, and reporting — with one unified, real-time connected platform.

Key Highlights

- **120+ API endpoints** powering complete distribution operations
- **20+ desktop screens** with full financial dashboards, aging reports, and PDF generation
- **Offline-first mobile app** with auto-sync — works in zero-connectivity zones
- **Multi-warehouse stock management** with real-time reservation and deduction
- **Double-entry shop ledger** with FIFO payment allocation and aging analysis
- **Thermal receipt & delivery challan PDF generation** for POS printers
- **Role-based access control** — Admin, Manager, Salesman, Warehouse
- **Two deployment modes** — Cloud-connected client or fully standalone offline package

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2. The Problem We Solve

The Distribution Industry's Pain Points

Distribution businesses, especially in emerging markets across South Asia, the Middle East, Africa, and Southeast Asia, face a **cascading chain of operational inefficiencies**:

Problem 1: Paper-Based Order Taking

Pain	Impact
Salesmen write orders on paper during field visits	Orders are lost, illegible, or entered incorrectly
Manual data entry at the office	2–4 hour delay before orders enter the system
No real-time visibility	Management has no idea what was sold until end of day

Our Solution: Mobile app with offline-first order taking. Orders are digitized at the shop counter, synced automatically when connectivity is available — zero paper, zero delay.

Problem 2: Disconnected Systems

Pain	Impact
Separate software for inventory, billing, delivery, and accounting	Data silos — stock levels don't match orders, invoices don't match deliveries
Excel spreadsheets for financial tracking	Errors compound, reconciliation takes days
No single source of truth	Conflicting data leads to disputes with shop owners

Our Solution: One unified platform where an order placed by a salesman automatically reserves stock, generates delivery challans, updates the shop ledger, and feeds the financial dashboard — all in real-time.

Problem 3: Stock Visibility Failures

Pain	Impact
No real-time stock levels	Salesmen promise items that aren't in stock
No multi-warehouse view	Stock exists but in the wrong warehouse
Over-ordering or under-ordering from suppliers	Dead stock or stockouts

Our Solution: Multi-warehouse stock management with reserved stock tracking, low-stock alerts, reorder level monitoring, and stock movement audit trails.

Problem 4: Cash Collection & Credit Risk

Pain	Impact
No credit limit enforcement	Shops accumulate dangerous levels of debt
Manual payment tracking	Lost payments, disputes, unreconciled accounts
No aging analysis	No visibility into overdue receivables

Our Solution: Built-in shop ledger with credit limit enforcement, FIFO payment allocation, aging analysis (current/30/60/90+ days), and real-time receivables dashboard.

Problem 5: Field Team Accountability

Pain	Impact
No way to verify if a salesman visited a shop	Ghost visits, inflated claims
No performance metrics	No data to reward top performers or coach underperformers
No target tracking	Salesmen have no visibility into their own progress

Our Solution: Digital order trail per salesman, monthly target tracking with progress bars, performance analytics, commission calculation, and route-based isolation.

Problem 6: Delivery Reconciliation Chaos

Pain	Impact
Deliveries tracked on paper	Partial deliveries, returns, and shortages go unrecorded
No challan system	Shops dispute what was delivered
No return processing	Returned stock disappears from records

Our Solution: Digital delivery challans with thermal-print PDF generation, partial delivery tracking, stock return processing, and load sheet management for warehouse operations.

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3. Product Overview

The Three Pillars

Pillar 1: Desktop Command Center (Windows)

The desktop application is the **operational hub** for managers and administrators. Built with Electron and React 19, it provides a rich, responsive interface for managing every aspect of the distribution business.

Core Capabilities:

- Real-time dashboard with KPI cards and auto-refresh (30-second intervals)
- Complete product catalog management with CSV bulk import
- Multi-warehouse inventory control with stock levels, movements, and alerts
- Order lifecycle management (Draft → Placed → Approved → Finalized → Delivered)
- Delivery challan generation and tracking
- Financial ledger system with aging reports and payment recording
- Daily collection tracking
- Stock return processing
- Route and salesman management with performance analytics
- Company settings and branding configuration
- Professional thermal receipt PDF generation (80mm format)
- Print-ready reports across all modules

Pillar 2: Mobile Field App (Android)

The mobile application is the **field salesman's daily companion**. Built with React Native and Expo SDK 54, it delivers a fast, intuitive experience designed for use in the field — even without internet connectivity.

Core Capabilities:

- Offline-first architecture with local SQLite database
- Complete order-taking workflow with product search, category filtering, and cart management
- Per-item and order-level discount application
- Shop listing with credit status indicators (green/amber/red)
- Monthly sales target tracking with visual progress
- Auto-sync with exponential backoff retry (works on network restoration)
- Background sync that continues even after app termination

- Direct phone call and SMS to shop owners
- Pull-to-refresh on all data screens
- Server connection testing and runtime configuration

Pillar 3: Cloud API Server

The backend server is the **data backbone** that connects desktop and mobile, enforces business rules, and persists all operational data.

Core Capabilities:

- RESTful API with 120+ endpoints across 22 route modules
- JWT authentication with session management
- Role-based authorization (Admin, Manager, Salesman, Warehouse)
- In-memory LRU cache for high-performance dashboard queries
- Rate limiting on mobile sync endpoints (100 requests/15 minutes)
- MySQL database with 30+ tables, views, and stored procedures
- Bidirectional sync engine for mobile devices
- Conflict detection and resolution for concurrent operations
- Stock reservation via stored procedures (thread-safe)
- Image upload and compression middleware

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4. Platform Architecture

```
┌──────────────────────────────────────────────────────────────────────┐
│                    DISTRIBUTION MANAGEMENT SYSTEM                    │
├──────────────────────────────────────────────────────────────────────┤
│                                                                      │
│   ┌──────────────────┐     ┌──────────────────┐                     │
│   │  DESKTOP APP     │     │  MOBILE APP      │                     │
│   │  (Windows)       │     │  (Android)       │                     │
│   │                  │     │                  │                     │
│   │  • Electron      │     │  • React Native  │                     │
│   │  • React 19      │     │  • Expo SDK 54   │                     │
│   │  • Tailwind CSS  │     │  • SQLite (WAL)  │                     │
│   │  • Material UI   │     │  • Paper UI      │                     │
│   │  • jsPDF         │     │  • Auto-Sync     │                     │
│   └──────────┬───────┘     └──────────┬───────┘                     │
│              │                         │                             │
│              │    HTTPS / REST         │                             │
│              └─────────────┬───────────┘                             │
│                            │                                         │
│              ┌─────────────▼─────────────┐                           │
│              │   BACKEND API             │                           │
│              │   (Node.js)               │                           │
│              │                           │                           │
│              │   • Express 5             │                           │
│              │   • JWT Auth              │                           │
```

```
│            │   • RBAC            │                                  │
│            │   • LRU Cache       │                                  │
│            │   • Rate Limiting   │                                  │
│            │   • 120+ Endpoints  │                                  │
│            └─────────────────────┘                                  │
│                       │                                             │
│            ┌──────────▼──────────┐                                  │
│            │   DATABASE          │                                  │
│            │   (MySQL 8.0+)      │                                  │
│            │                     │                                  │
│            │   • 30+ Tables      │                                  │
│            │   • Views           │                                  │
│            │   • Stored Procs    │                                  │
│            │   • Audit Trails    │                                  │
│            └─────────────────────┘                                  │
│                                                                     │
└─────────────────────────────────────────────────────────────────────┘
```

Data Flow

```
SALESMAN IN FIELD                    OFFICE / HEADQUARTERS
────────────────                     ─────────────────────

1. Visit Shop                         5. Manager reviews orders
2. Take Order (offline)                  on Dashboard
3. Order saved to SQLite              6. Approve/Reject orders
4. Auto-sync when online   ──►       7. Finalize → stock deducted
                                      8. Generate delivery challan
                                      9. Dispatch from warehouse
                                     10. Record delivery & returns
                                     11. Update shop ledger
                                     12. Collect payments
                                     13. Monitor aging & receivables
```

5. Comprehensive Feature Breakdown

5.1 Dashboard & Business Intelligence

Feature	Description
Real-Time KPI Cards	Live counts of products, orders, deliveries, shops, salesmen, warehouses
Auto-Refresh	Dashboard data refreshes every 30 seconds automatically
Revenue Summary	Total revenue, daily/weekly/monthly breakdowns
Sales Trends	Historical sales data visualization

Feature	Description
City-Wise Statistics	Geographic performance breakdown
Salesman Target Progress	Individual target achievement tracking
Top Performers	Ranked lists of top salesmen and best-selling products
Low Stock Alerts	Immediate visibility into products needing reorder
Quick Action Shortcuts	One-click access to frequent operations

5.2 Product & Catalog Management

Feature	Description
Full CRUD Operations	Create, read, update, soft-delete products
Multi-Attribute Catalog	Category, brand, company, unit type, barcode support
Multi-Price Tiers	Unit price, carton price, purchase price, wholesale price
CSV Bulk Import	Upload hundreds of products via spreadsheet with preview validation
Advanced Filtering	Filter by category, brand, company, stock level, active status
Paginated Listings	Configurable page sizes with search across all fields
Supplier Linkage	Every product linked to its supplier for procurement tracking
Profit Margin Calculation	Automatic margin tracking based on purchase vs. selling price
Barcode Support	Schema-ready for barcode scanning integration

5.3 Multi-Warehouse Inventory Management

Feature	Description
Multiple Warehouses	Manage stock across unlimited warehouse locations
Default Warehouse	Designate a primary warehouse for order fulfillment
Per-Warehouse Stock Levels	Individual quantity tracking per product per warehouse
Stock Reservation	Automatic hold on stock when orders are placed
Stock Deduction	Automatic deduction when orders are finalized (via stored procedures)
Low-Stock & Reorder Alerts	Configurable minimum stock and reorder levels
Stock Movement Audit	Complete trail of every stock change with before/after quantities
Batch & Expiry Tracking	Track batch numbers and expiry dates per warehouse stock entry
Capacity Monitoring	Warehouse capacity tracking and utilization
Storage Type Classification	Categorize warehouses (ambient, cold, dry, etc.)

5.4 Order Management System

Feature	Description
Multi-Channel Order Creation	Orders from mobile (field) and desktop (office)
Smart Order Numbering	Auto-generated format: ORD-YYYYMMDD-SXXX-NNNNN
Order Lifecycle	Draft → Placed → Approved → Finalized → Delivered (or Rejected)
Stock Availability Check	Real-time stock verification before order approval
Discount System	Per-item percentage discounts + order-level fixed discounts
Order Statistics Dashboard	Aggregate metrics across all orders
Pending Order Queue	Quick view of orders awaiting action
Order History	Historical record with date range and status filters
Approval Workflow	Manager review and approval before stock commitment
Bulk Operations	Multi-select operations for efficiency

5.5 Delivery & Logistics

Feature	Description
Delivery Challan Generation	Professional challans (DC-YYYYMMDD-XXXX) from approved orders
Driver & Vehicle Tracking	Record driver name, vehicle number, departure/arrival times
Status Lifecycle	Pending → In Transit → Delivered → Cancelled
Partial Delivery Support	Record actual quantities delivered vs. ordered
Thermal Receipt PDFs	80mm thermal-print optimized PDF generation
Load Sheet Management	Consolidate products across multiple deliveries for warehouse loading
Bulk Operations	Bulk status update and delete capabilities
Delivery Statistics	Performance metrics across all deliveries
Receiver Confirmation	Record receiver name at point of delivery

5.6 Financial Management & Shop Ledger

Feature	Description
Double-Entry Ledger	Every transaction creates matching debit/credit entries
Running Balance Calculation	Automatic real-time balance per shop
Credit Limit Management	Set and enforce per-shop credit limits
Credit Risk Indicators	Color-coded risk assessment (Green/Amber/Red)

Feature	Description
FIFO Payment Allocation	Payments automatically allocated to oldest outstanding invoices
Aging Analysis	Bucketed receivables: Current / 1-30 / 31-60 / 61-90 / 90+ days
Financial Dashboard	Total receivables, risk metrics, critical alerts, top debtors
Payment Recording	Record payments with receipt numbers (RCP-YYYYMMDD-XXXX)
Multiple Payment Methods	Cash, cheque (with bank details), bank transfer
Manual Adjustments	Admin-authorized balance corrections with audit trail
Account Statements	Printable per-shop transaction history
Advance Payment Handling	Process and track advance payments from shops
Balance Recalculation	Admin tool to recalculate and verify running balances

5.7 Route & Territory Management

Feature	Description
Route Definition	Create routes with code, name, area, and city
Salesman Assignment	Assign routes to specific salesmen
Route Statistics	Performance metrics per route
Consolidated Route Bills	Aggregate billing report per route with date range selection
Multi-Route Support	Salesmen can manage multiple routes
Printable Route Reports	Print-ready consolidated bill format

5.8 Shop & Customer Management

Feature	Description
Comprehensive Shop Profiles	Name, owner, address, contact, GPS coordinates
Shop Type Classification	Categorize shops (retail, wholesale, etc.)
Credit Limit Configuration	Per-shop credit ceiling
Tax Information	GST/NTN registration tracking per shop
Route Assignment	Each shop linked to its delivery route
Contact Integration	Direct call and SMS from mobile (one-tap)
Search & Filter	Find shops by name, owner, phone, shop code
Printable Shop Directory	Print-ready shop listing

5.9 Salesman Management & Performance

Feature	Description
Salesman Profiles	Name, code, CNIC, phone, vehicle type, address
Monthly Targets	Set revenue targets per salesman
Commission Tracking	Configurable commission percentage
Performance Analytics	Order counts, revenue, target achievement
Credential Management	Create login accounts, reset passwords
Salary Ledger	Track salary, advances, bonuses, deductions
Route Assignment	Assign/unassign routes to salesmen
Active/Inactive Status	Soft-delete with deactivation

5.10 Stock Returns Management

Feature	Description
Return Processing	Process returns against specific delivery challans
Return Numbering	Auto-generated: RET-YYYYMMDD-XXXX
Item-Level Returns	Select specific items and quantities to return
Reason Tracking	Record return reasons and notes
Stock Restoration	Returned stock automatically added back to warehouse
Return Statistics	Aggregate return metrics and trends

5.11 Daily Collections

Feature	Description
Collection Recording	Record daily cash and payment collections
Payment Method Tracking	Cash, cheque, bank transfer categorization
Today's Summary	Real-time view of today's collections
Historical Reports	Date range queries with aggregated summaries
Shop & Salesman Linkage	Track who collected from which shop

5.12 Mobile Offline-First Capabilities

Feature	Description
Local SQLite Database	Full data replica on device with WAL mode
Zero-Connectivity Operation	Complete order-taking without any internet
Automatic Background Sync	Syncs when connectivity restores, even after app kill

Feature	Description
Exponential Backoff Retry	Smart retry: 1min → 5min → 15min → 30min → 60min
Connectivity Banner	Visual online/offline indicator
Sync Status per Order	Clear indicators: synced vs. waiting to sync
Pull-to-Refresh	Manual sync trigger on any screen
Database Optimization	Memory-mapped I/O, prepared statements, 20+ indexes
Query Result Caching	5-minute TTL cache for repeated queries
Batch Sync	Upload up to 50 orders in a single sync operation
Sync Audit Log	Last 50 sync attempts logged with timestamps and results

5.13 Reporting & Document Generation

Feature	Description
Thermal Receipt PDFs	80mm POS-optimized delivery challans
Printable Reports	Direct browser print across 7+ screens
Aging Reports	Bucketed receivables with totals and percentages
Consolidated Route Bills	Per-route product and shop level breakdown
Account Statements	Per-shop transaction history with running balance
Order Statistics	Aggregate order metrics with filters
Delivery Statistics	Logistics performance reporting
Return Statistics	Return trend analysis
Dashboard Analytics	Real-time business intelligence

5.14 Company Settings & Branding

Feature	Description
Company Profile	Name, address, phone, email, website
Tax Configuration	GST/tax rate, NTN number
Bank Details	Bank name, account title, account number, IBAN
Branding	Logo upload for invoices and challans
Currency Settings	Configurable currency symbol and formatting
Invoice Customization	Terms and conditions, notes

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6. User Roles & Access Control

The system implements a robust **Role-Based Access Control (RBAC)** framework with four distinct user roles:

Role	Desktop Access	Mobile Access	Key Permissions
Admin	Full access to all 20+ screens	—	Company settings, user management, ledger clearing, product deletion, manual adjustments
Manager	All operational screens	—	Order approval/rejection, delivery management, payment recording, stock management
Salesman	—	Full mobile app access	Order creation, shop browsing, product catalog, target tracking
Warehouse	Warehouse-specific screens	—	Stock management, load sheets, dispatching

Permission Hierarchy

```
Admin (Full Control)
  └── Manager (Operations)
        └── Salesman (Field Operations)
        └── Warehouse (Stock Operations)
```

Security Features per Role

- **Admin**: Can delete records, clear histories, adjust balances, manage users
- **Manager**: Can approve orders, manage deliveries, record payments — cannot delete critical data
- **Salesman**: Can only see their assigned route's shops, cannot see purchase prices or profit margins
- **Warehouse**: Manages stock levels and dispatching within their assigned warehouse

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7. Deployment Models

The DMS offers **four flexible deployment configurations** to suit any business size and infrastructure:

Model A: Cloud-Hosted (Recommended for Multi-Location)

```
┌──────────────┐      ┌──────────────┐
│ Desktop App  │──►│  VPS/Cloud   │◄──│ Mobile App │
│ (Office 1)   │      │  Server      │      │ (Field)    │
└──────────────┘      │  + MySQL     │      └──────────────┘
                      │  + Nginx     │
┌──────────────┐      │  + SSL       │
│ Desktop App  │──►│
```

```
│ (Office 2)   │      └──────────────┘
└──────────────┘
```

Best for: Multi-branch operations, remote management **Requirements:** VPS (2GB RAM, 20GB storage), domain name, SSL certificate

Model B: Standalone Desktop (Single Location)

```
┌──────────────────────────────┐
│  Desktop App                 │
│  ┌──────────┐ ┌──────────┐   │
│  │ Frontend │ │ Backend  │   │
│  │ (React)  │ │ (.exe)   │   │
│  └──────────┘ └──────────┘   │
│                  │           │
│               ┌──▼──┐        │
│               │SQLite│       │
│               └─────┘        │
└──────────────────────────────┘
```

Best for: Single-location, small businesses, no IT staff **Requirements:** Windows PC only — nothing else needed

Model C: LAN Network (Office-Based)

```
┌──────────────┐     ┌──────────────┐
│ Desktop App  │────►│ Server PC    │◄────│ Desktop App │
│ (PC 1)       │     │ (Backend)    │     │ (PC 2)      │
└──────────────┘     └──────────────┘     └──────────────┘
```

Best for: Offices without internet, multiple desktop users **Requirements:** Windows PCs on same network

Model D: Hybrid (Cloud + Standalone)

Best for: Businesses wanting cloud benefits with offline resilience **Configuration:** Cloud server as primary, standalone mode as fallback

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8. Target Audience & Ideal Customer Profile

8.1 Primary Target Segments

Segment 1: FMCG Distributors

Attribute	Profile

Attribute	Profile
Business Size	50–500 shops in distribution network
Team Size	5–50 salesmen in the field
Products	Food, beverages, snacks, household goods
Pain Point	Paper-based ordering, stock mismatches, cash collection delays
Decision Maker	Owner / General Manager
Budget Range	$500–$5,000 one-time or $50–$500/month

Segment 2: Pharmaceutical Distributors

Attribute	Profile
Business Size	100–1,000 pharmacies/medical stores
Team Size	10–100 field representatives
Products	Medicines, medical supplies, healthcare products
Pain Point	Batch/expiry tracking, regulatory compliance, returns management
Decision Maker	Distribution Manager / Owner
Budget Range	$1,000–$10,000 one-time or $100–$1,000/month

Segment 3: Building Materials Distributors

Attribute	Profile
Business Size	30–300 retail shops / hardware stores
Team Size	3–30 salesmen
Products	Cement, paint, tiles, plumbing, electrical
Pain Point	Heavy credit-based sales, aging receivables, delivery tracking
Decision Maker	Owner / Operations Manager
Budget Range	$500–$5,000

Segment 4: Cosmetics & Personal Care Distributors

Attribute	Profile
Business Size	100–500 beauty shops and parlors
Team Size	5–30 salesmen
Products	Skincare, haircare, fragrances, makeup
Pain Point	Brand-wise tracking, discount management, route optimization

Attribute	Profile
Decision Maker	Sales Director / Owner

Segment 5: Agricultural Input Distributors

Attribute	Profile
Business Size	50–200 rural retail points
Team Size	5–20 salesmen covering rural areas
Products	Seeds, fertilizers, pesticides, tools
Pain Point	Seasonal demand, credit management, remote areas with no connectivity
Decision Maker	Regional Manager / Owner

Segment 6: Stationery & Office Supplies Distributors

Attribute	Profile
Business Size	100–500 bookshops / office supply stores
Team Size	5–25 salesmen
Products	Notebooks, pens, office supplies, school supplies
Pain Point	Seasonal demand cycles, large SKU catalogs, bulk order management

8.2 Secondary Target Segments

Segment	Description
Beverage Distributors	Soft drinks, juices, water — high frequency, route-based
Tobacco Distributors	Cigarettes, tobacco products — strict route compliance
Confectionery Distributors	Chocolates, candies, biscuits — high SKU count
Electronics Accessories	Phone accessories, cables — high volume, low value
Auto Parts Distributors	Vehicle spare parts — credit-heavy, returns common
Textile & Garment Wholesalers	Fabric, ready-made garments — seasonal, bulk orders

8.3 Decision Maker Profiles

Role	Concerns	Value Proposition
Business Owner	ROI, cost savings, control	"See your entire business on one screen"
General Manager	Operational efficiency, team accountability	"Know exactly what every salesman is doing, in real-time"

Role	Concerns	Value Proposition
Sales Director	Revenue growth, target tracking	"Set targets, track progress, reward top performers"
Finance Manager	Receivables, cash flow, collections	"Never lose track of who owes you what"
IT Manager	Deployment ease, security, maintenance	"One installer, no complex setup, secure by default"

9. Industry Applications & Use Cases

Use Case 1: The FMCG Distributor

Scenario: A snack food distributor manages 200 shops across 15 routes with 10 salesmen.

Before DMS:

- Salesmen carry paper order books; 15% of orders have errors
- Office staff spend 3 hours daily entering orders into Excel
- Stock discrepancies discovered only during monthly audits
- ₹2.5M in unpaid receivables with no aging visibility
- Delivery disputes occur weekly due to lack of documentation

After DMS:

- Orders digitized at the shop — zero paper, zero data entry
- Real-time stock visibility across 2 warehouses
- ₹800K in receivables recovered within 60 days via aging reports
- 99% reduction in delivery disputes with digital challans
- Salesmen achieve 30% more shop visits per day
- Management makes data-driven decisions from the dashboard

Use Case 2: The Pharmaceutical Distributor

Scenario: A pharma distributor serves 500 pharmacies with 25 field representatives.

After DMS:

- Batch and expiry tracking prevents expired product delivery
- Credit limit enforcement prevents risky pharmacy accounts from growing
- Returns processed digitally with reason tracking for manufacturer credits
- FIFO payment allocation ensures oldest invoices are settled first
- Regulatory audit trail maintained automatically

Use Case 3: The Rural Distributor

Scenario: An agricultural inputs distributor serves 100 shops in remote areas with poor connectivity.

After DMS:

- Offline-first mobile app works perfectly in zero-signal rural areas
- Orders taken offline, synced automatically when salesman returns to coverage area
- Background sync continues even when app is closed
- No lost orders — every transaction is persisted locally first
- Route-based shop assignment ensures territory coverage

Use Case 4: The Multi-Branch Operation

Scenario: A consumer goods company operates 3 warehouses in different cities.

After DMS:

- Cloud-hosted server connects all 3 branches in real-time
- Per-warehouse stock visibility from any location
- Consolidated financial dashboard spans all operations
- Load sheet management for each warehouse independently
- Centralized order approval with branch-level fulfillment

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10. Global Market Opportunity

10.1 Target Regions

Tier 1: Primary Markets (Immediate)

Region	Countries	Market Drivers
South Asia	Pakistan, India, Bangladesh, Sri Lanka	Massive distribution networks, paper-based operations, growing digitization
Middle East	UAE, Saudi Arabia, Qatar, Oman, Kuwait, Bahrain	Wholesale distribution culture, high smartphone penetration, trade route networks
Southeast Asia	Indonesia, Philippines, Vietnam, Thailand, Malaysia	FMCG distribution boom, emerging digital infrastructure

Tier 2: High-Growth Markets

Region	Countries	Market Drivers
East Africa	Kenya, Tanzania, Ethiopia, Uganda	Rapid FMCG growth, mobile-first infrastructure, distribution challenges
West Africa	Nigeria, Ghana, Senegal, Côte d'Ivoire	Largest consumer markets in Africa, distribution gap
North Africa	Egypt, Morocco, Tunisia, Algeria	Established trade networks, digital transformation underway

Region	Countries	Market Drivers
Central Asia	Uzbekistan, Kazakhstan, Kyrgyzstan	Emerging markets, distribution infrastructure being built

Tier 3: Expansion Markets

Region	Countries	Market Drivers
Latin America	Mexico, Colombia, Peru, Brazil	Large distribution networks, mobile adoption
Eastern Europe	Turkey, Romania, Poland, Ukraine	Growing wholesale/distribution sector
South Pacific	Papua New Guinea, Fiji, Pacific Islands	Remote distribution challenges (offline-first appeal)

10.2 Market Size & Opportunity

Metric	Value
Global Distribution Software Market	$4.2 billion (2025), growing at 11.2% CAGR
FMCG Distribution Market (Asia-Pacific)	$1.8 trillion in goods distributed annually
Number of Distributors (India alone)	7+ million registered distributors
Number of Distributors (Pakistan)	500,000+ active distribution businesses
Mobile Penetration (target markets)	70–95% smartphone ownership
Digital Adoption Gap	Less than 15% of distributors in emerging markets use specialized software

10.3 Why Now?

1. **Post-COVID Digital Acceleration** — Businesses that resisted digitization are now actively seeking solutions
2. **Smartphone Ubiquity** — Even rural salesmen carry smartphones capable of running the mobile app
3. **Cloud Infrastructure Growth** — Affordable VPS hosting available globally ($5–20/month)
4. **Competition Gap** — Enterprise solutions (SAP, Oracle) are priced out of reach for SME distributors
5. **Government Digitization Mandates** — Tax authorities increasingly requiring digital records

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11. Competitive Advantages & USPs

11.1 Unique Selling Propositions

USP 1: True Offline-First Architecture

> **"The only distribution app that works with zero internet — not just 'offline mode', but offline by design."**

Unlike competitors that simply cache data, our mobile app has a **complete local SQLite database** with full order-taking capability. Orders are created, stored, and managed locally. Sync happens opportunistically — not as a requirement.

- Background sync runs even after the app is terminated
- Exponential backoff retry prevents battery drain
- 50-order batch upload for efficient bandwidth use
- Sync audit log for complete visibility

USP 2: All-in-One Platform (5-in-1)

> **"Replace your entire software stack with one system."**

Replaces	Module in DMS
Order Management Software	Order Management System
Warehouse Management (WMS)	Multi-Warehouse Inventory
Accounting/Ledger Software	Shop Ledger & Financial Dashboard
Delivery/Logistics Software	Delivery Challan & Load Sheet Management
Field Sales App	Mobile App for Salesmen
Billing/Invoice Software	Invoice & Payment System
Reporting/BI Tool	Dashboard & Analytics

USP 3: Standalone Desktop Mode

> **"One installer. No internet. No server. No IT department. Just download and run."**

The standalone build bundles the entire backend inside the desktop application. For small businesses, there is **zero infrastructure requirement** — just install on a Windows PC and start operating.

USP 4: Financial Intelligence Built-In

> **"Know who owes you money, how much, and for how long — in seconds."**

The double-entry ledger with FIFO payment allocation and aging analysis provides financial visibility that typically requires separate accounting software:

- Current / 30-day / 60-day / 90-day / 90+ day aging buckets
- Credit limit enforcement prevents risky sales
- Real-time receivables dashboard with risk alerts
- Printable account statements for shop owners

USP 5: Zero Training Order-Taking

> **"Your salesman can take their first digital order in under 2 minutes."**

The mobile app follows a natural workflow: **Select Shop** → **Browse Products** → **Add to Cart** → **Submit**. The interface uses Material Design 3 with large touch targets, visual status indicators, and haptic feedback — designed for field use, not IT professionals.

USP 6: Thermal Printer Integration

> **"Professional delivery challans printed right at the shop doorstep."**

Built-in PDF generation optimized for 80mm thermal receipt printers — the same printers already used by most distribution businesses. No additional software or drivers needed.

11.2 Competitive Comparison

Feature	Our DMS	SAP B1 / Oracle	Local ERP Tools	Excel/Paper
Price	Affordable	$50K–500K+	$1K–10K	Free
Setup Time	Hours	Months	Weeks	Immediate
Offline Mobile	Full offline	Limited/None	Rare	N/A
Desktop App	Native Windows	Web only	Variable	N/A
Financial Ledger	Built-in	Separate module	Basic	Manual
Multi-Warehouse	Built-in	Add-on	Rare	Manual
Delivery Challans	Auto-generated PDF	Custom dev	Manual	Paper
Salesman Tracking	Built-in	Custom dev	Limited	None
Learning Curve	1–2 days	3–6 months	2–4 weeks	None
IT Staff Required	No	Yes (team)	Sometimes	No
Customization	Configurable	Costly	Limited	Unlimited

12. Security & Compliance

Authentication & Authorization

Feature	Implementation
Password Hashing	bcrypt with salt rounds (industry standard)
Token Authentication	JWT with configurable expiry (default 7 days)
Session Management	Server-side session validation for every request
Role-Based Access	Four-tier RBAC: Admin > Manager > Salesman > Warehouse
Account Deactivation	Immediate lockout via `is_active` flag

Feature	Implementation
Password Reset	Admin-controlled password reset for salesmen

Data Security

Feature	Implementation
SQL Injection Prevention	Parameterized queries throughout (mysql2 prepared statements)
XSS Prevention	Input sanitization and output encoding
Rate Limiting	Mobile sync: 100 requests per 15 minutes per IP
CORS Configuration	Configurable origin restrictions
HTTPS Support	SSL/TLS encryption via Nginx reverse proxy
Context Isolation	Electron app runs with context isolation enabled

Data Privacy

Feature	Implementation
Data Locality	Self-hosted option keeps all data on client's own infrastructure
No Third-Party Data Sharing	Zero external analytics or tracking; data stays internal
Soft Delete	Records are deactivated, not destroyed — preserving audit trails
Audit Trails	Stock movements, sync logs, ledger entries all timestamped

Compliance Readiness

Standard	Status
Tax Record Keeping	GST/NTN fields on shops, tax rates configurable
Financial Audit Trail	Double-entry ledger with immutable transaction history
Stock Audit Trail	Complete before/after record of every stock change
User Activity Logging	Session-based authentication with login tracking

13. Return on Investment (ROI)

Quantifiable Business Impact

Time Savings

Activity	Before DMS	After DMS	Savings
Order data entry (per day)	3–4 hours	0 hours (automated)	~780 hours/year

Activity	Before DMS	After DMS	Savings
Stock reconciliation	8 hours/month	30 min/month	~90 hours/year
Delivery documentation	2 hours/day	15 min/day	~430 hours/year
Payment reconciliation	4 hours/week	30 min/week	~180 hours/year
Total Time Saved			**~1,480 hours/year**

Revenue Impact

Metric	Impact
Order Accuracy	99%+ (vs. 85% with paper) — fewer disputes, fewer returns
Salesman Productivity	20–30% more shop visits per day (no paperwork at each stop)
Receivables Recovery	15–25% faster collection via aging visibility and credit enforcement
Stock Waste Reduction	10–20% reduction via real-time visibility and reorder alerts
Delivery Dispute Reduction	95%+ reduction with digital challan evidence

Cost Savings

Cost Center	Annual Savings (Typical 200-shop Distributor)
Data entry staff (1–2 FTEs)	$3,000–$8,000
Paper, printing, order books	$500–$1,500
Lost/duplicate orders	$2,000–$5,000
Uncollected receivables (improved by 15%)	$5,000–$25,000
Stock wastage (improved by 15%)	$2,000–$10,000
Total Annual Savings	**$12,500–$49,500**

ROI Calculation

Scenario	Investment	Annual Savings	ROI	Payback Period
Small (50 shops)	$500–$1,500	$5,000–$12,000	700%+	< 2 months
Medium (200 shops)	$1,500–$5,000	$12,500–$49,500	500%+	< 2 months
Large (500+ shops)	$5,000–$15,000	$30,000–$100,000+	400%+	< 3 months

14. Marketing Strategy & Go-To-Market Plan

14.1 Digital Marketing Channels

Search Engine Marketing (SEM)

High-Intent Keywords to Target:

Keyword Category	Examples
Problem-Based	"distribution management software", "sales order tracking system", "FMCG distribution solution"
Role-Based	"salesman order taking app", "distributor accounting software", "field sales management"
Industry-Based	"pharma distribution software", "FMCG distribution app", "wholesale management system"
Region-Based	"distribution software Pakistan", "sales management app India", "wholesale ERP UAE"
Comparison	"SAP alternative for distributors", "affordable distribution ERP", "best distribution management app"

Content Marketing Strategy

Content Type	Topics	Frequency
Blog Posts	"5 Signs Your Distribution Business Needs Software", "How Offline-First Apps Save Rural Distributors"	2–4/month
Case Studies	Before/after stories from each industry vertical	1/month
Whitepapers	"The True Cost of Paper-Based Distribution", "Credit Risk Management for Distributors"	1/quarter
Video Demos	Screen-by-screen walkthrough, mobile app demo, installation guide	Library of 10+
Webinars	"Digitize Your Distribution in 7 Days" — live demonstration events	1–2/month
Infographics	ROI calculator, feature comparison charts, workflow diagrams	2/month

Social Media Strategy

Platform	Audience	Content Focus
LinkedIn	Business owners, managers, IT decision-makers	Thought leadership, case studies, ROI data
Facebook	Small business owners in South Asia, Middle East, Africa	Product demos, success stories, community building
YouTube	Technical buyers, implementation teams	Full demo videos, installation tutorials, tips & tricks

Platform	Audience	Content Focus
Instagram	Brand awareness in younger demographics	Behind-the-scenes, quick feature highlights, infographics
WhatsApp Business	Direct engagement in key markets	Quick demos, support, lead nurturing

14.2 Direct Sales & Channel Strategy

Direct Sales Approach

Stage	Activity
Lead Generation	Industry directories, trade show lists, LinkedIn outreach, Google Ads
First Contact	Free demo scheduling via website, WhatsApp, or phone
Live Demo	30-minute personalized demo showing the prospect's industry workflow
Trial Period	14–30 day free trial with sample data pre-loaded
Objection Handling	ROI calculator, competitor comparison, reference customers
Closing	Flexible payment terms, installation support included
Onboarding	Data migration assistance, training sessions, 30-day support

Channel Partner Strategy

Partner Type	Value to Us	Value to Partner
IT Resellers	Local distribution and support	Recurring revenue, expanded service offering
Business Consultants	Trusted advisor recommendations	Solution to offer consulting clients
Industry Associations	Access to member directories, event sponsorship	Digital transformation offering for members
POS Hardware Vendors	Bundled sales with thermal printers	Software that drives hardware sales
Cloud Hosting Providers	Infrastructure referrals	Value-added service for hosting clients

14.3 Industry-Specific Marketing

FMCG Distributors

- **Message:** "From paper order books to digital orders in one day"
- **Channels:** Industry trade shows, FMCG WhatsApp groups, distributor associations
- **Proof Points:** Order accuracy rates, time savings, stock waste reduction

Pharmaceutical Distributors

- **Message:** "Batch tracking, expiry management, and regulatory-ready records"
- **Channels:** Pharma distribution conferences, medical trade publications
- **Proof Points:** Compliance audit trail, returns management, credit controls

Building Materials Distributors

- **Message:** "Stop losing money on overdue credit — see your aging in real-time"
- **Channels:** Construction industry events, hardware trade groups
- **Proof Points:** Receivables recovery, credit limit enforcement, collection improvement

14.4 Offline Marketing Tactics

Tactic	Approach
Trade Shows & Exhibitions	Demo booth with live system showcase, thermal printer printing challans on the spot
Distributor Association Events	Sponsored presentations on digital transformation
Referral Program	Existing clients earn credits for every referral that converts
Door-to-Door Sales	Visit distribution markets in major cities with tablets showing live demos
Newspaper/Magazine Ads	Trade publications in target industries
Radio Advertising	Local business radio in key markets (Pakistan, India, Nigeria)
Branded Merchandise	Thermal receipt rolls with DMS branding, pens, notebooks for distributors

14.5 Strategic Partnerships

Partner Category	Partnership Model
Telecom Companies	Bundle mobile app with business data plans
Banks	Integration with payment gateways; banks recommend to lending clients
Government Programs	Partner with SME digitization initiatives
Accounting Firms	Recommend to auditing clients for better financial records
ERP Consultants	Position as the affordable ERP for distribution SMEs

15. Pricing Strategy Framework

15.1 Pricing Models

Model A: One-Time License

Tier	Users	Features	Price Range
Starter	1 Admin + 3 Salesmen	Core features, 1 warehouse, standalone mode	$299–$499
Professional	1 Admin + 2 Managers + 10 Salesmen	All features, 3 warehouses, cloud deployment	$799–$1,499
Enterprise	Unlimited users	All features, unlimited warehouses, multi-branch, priority support	$2,499–$4,999

Model B: Monthly Subscription (SaaS)

Tier	Users	Features	Monthly Price
Basic	Up to 5 users	Core features, 50 shops, 1 warehouse	$29–$49/month
Growth	Up to 15 users	All features, 200 shops, 3 warehouses	$79–$149/month
Scale	Up to 50 users	All features, unlimited shops/warehouses	$199–$399/month
Enterprise	Unlimited	Custom features, dedicated support, SLA	Custom pricing

Model C: Per-Salesman Pricing

Component	Price
Base Platform (Admin + Management)	$99–$199/month
Per Salesman License (Mobile app)	$10–$25/month per salesman
Example: 10 salesmen	$199 + (10 × $15) = $349/month

15.2 Add-On Revenue Streams

Add-On	Price Range
Installation & Setup	$100–$500 (one-time)
Data Migration (from Excel/existing systems)	$200–$1,000 (one-time)
Custom Training (on-site / remote)	$50–$200/session
Priority Support (24/7 response)	$50–$200/month
Custom Feature Development	$50–$150/hour
Thermal Printer Bundle	$80–$150 (hardware + setup)
Annual Maintenance Contract	15–20% of license fee

15.3 Free Tier / Freemium Strategy

Offering	Purpose
14-Day Free Trial	Full-featured trial to experience the system
Free Starter (1 user, 10 shops)	Low-barrier entry; upsell when business grows
Free Demo Instance	Pre-loaded sample data for evaluation
Free ROI Calculator	Web tool that calculates savings — lead generation magnet

16. Scalability & Future Roadmap

16.1 Current Scalability

Dimension	Current Capability
Users	Unlimited concurrent users (server-limited)
Products	Tested with 10,000+ SKUs
Shops	Tested with 1,000+ shops
Orders/Day	Handles 500+ orders/day
Warehouses	Unlimited warehouse locations
Database	MySQL 8.0 — scales to millions of records
Server	Horizontal scaling via load balancer + multiple API instances

16.2 Planned Feature Roadmap

Phase 1: Enhancement (Near-Term)

Feature	Business Value
GPS Location Tracking	Verify salesman shop visits with geofencing
Barcode Scanner Integration	Scan product barcodes for faster order entry
Push Notifications	Real-time alerts for order approvals, stock alerts
Photo Capture	Capture shop photos, product shelf images during visits
WhatsApp Integration	Send invoices and challans via WhatsApp directly

Phase 2: Intelligence (Mid-Term)

Feature	Business Value
AI Demand Forecasting	Predict order quantities based on historical patterns
Route Optimization	AI-suggested optimal visiting sequence

Feature	Business Value
Smart Reorder Suggestions	Automatic purchase order generation when stock is low
Customer Segmentation	AI-driven shop classification for targeted promotions
Anomaly Detection	Flag unusual order patterns, potential fraud

Phase 3: Expansion (Long-Term)

Feature	Business Value
iOS Mobile App	Expand to Apple ecosystem
Web Dashboard	Browser-based access for executives on any device
Multi-Company Support	Manage multiple distribution companies from one system
E-Commerce Integration	Connect with Shopify, WooCommerce for online orders
Payment Gateway Integration	Digital payments via JazzCash, EasyPaisa, UPI, M-Pesa
API Marketplace	Third-party integrations (accounting, logistics, CRM)
Multi-Language Support	Urdu, Arabic, Hindi, Bahasa, French, Swahili
Multi-Currency Support	INR, AED, USD, NGN with automatic conversion

17. Technical Specifications

17.1 Technology Stack

Component	Technology	Why It Matters
Backend Runtime	Node.js 18+	Fast, scalable, JavaScript ecosystem
API Framework	Express 5.1	Industry standard, battle-tested
Production Database	MySQL 8.0+	Enterprise-grade, ACID compliant, globally supported
Development Database	SQLite (better-sqlite3)	Zero-config development, portable
Mobile Database	expo-sqlite (WAL mode)	Optimized for mobile, concurrent reads
Authentication	JWT + bcrypt	Stateless, secure, scalable
Desktop Framework	Electron 39	Native Windows experience, auto-updates
Desktop UI	React 19 + Tailwind CSS + Material UI 7	Modern, responsive, accessible

Component	Technology	Why It Matters
Mobile Framework	React Native (Expo SDK 54)	Cross-platform, native performance
Mobile UI	React Native Paper (Material Design 3)	Native look and feel
PDF Generation	jsPDF + jspdf-autotable	Client-side PDF, no server dependency
Image Processing	Sharp	Efficient compression, thumbnail generation
Caching	In-memory LRU (100 entries, 5-min TTL)	Sub-millisecond dashboard queries
Process Manager	PM2 (production)	Zero-downtime restarts, monitoring
Reverse Proxy	Nginx	SSL termination, static serving, load balancing

17.2 System Requirements

Server (Cloud/VPS)

Requirement	Minimum	Recommended
OS	Ubuntu 20.04 LTS	Ubuntu 22.04 LTS
RAM	2 GB	4 GB
Storage	20 GB SSD	50 GB SSD
CPU	1 vCPU	2 vCPU
Network	100 Mbps	1 Gbps
Cost	~$10/month	~$20/month

Desktop Application

Requirement	Specification
OS	Windows 10 / 11 (64-bit)
RAM	4 GB minimum
Storage	500 MB for installation
Display	1280×720 minimum
Network	Required for cloud mode; not needed for standalone

Mobile Application

Requirement	Specification
OS	Android 10+

Requirement	Specification
RAM	2 GB minimum
Storage	100 MB + data cache
Network	Works fully offline; network for sync only

17.3 Database Architecture

Metric	Detail
Tables	30+ production tables
Views	4 materialized views for dashboard performance
Stored Procedures	3 critical stock operations (reserve, release, deduct)
Indexes	50+ indexes for query optimization
Migrations	20+ incremental migration scripts
Referential Integrity	Full foreign key constraints with CASCADE rules
Soft Deletes	`is_active` flag pattern — data never permanently lost

18. Success Metrics & KPIs

For Sales & Marketing Team

KPI	Target	Measurement
Demo Requests	50+/month within 6 months	Website form, WhatsApp inquiries
Trial Conversions	30%+ trial-to-paid rate	CRM tracking
Customer Acquisition Cost (CAC)	< $200 per client	Marketing spend / new clients
Monthly Recurring Revenue (MRR)	Growing 20%+ month-over-month	Subscription billing
Churn Rate	< 5% monthly	Active vs. churned clients
Net Promoter Score (NPS)	50+	Quarterly surveys
Average Contract Value (ACV)	$1,500+ (one-time) or $150+/month	Sales records

For Client Success (Post-Sale)

KPI	Target	Measurement

KPI	Target	Measurement
Onboarding Time	< 7 days from purchase to live	Project tracking
Daily Active Users	80%+ of licensed users	Login analytics
Orders Processed via App	90%+ of all orders digital within 30 days	System analytics
Support Ticket Resolution	< 24-hour average response	Helpdesk tracking
Feature Adoption Rate	60%+ using ledger, deliveries, returns	Usage analytics

19. Client Onboarding Process

Step-by-Step Implementation

```
Day 1-2: DISCOVERY & SETUP
├── Understand client's business workflow
├── Install desktop application (or provision cloud server)
├── Configure company settings and branding
└── Set up warehouses, routes, and user accounts

Day 3-4: DATA MIGRATION
├── Import product catalog (CSV bulk import)
├── Create shop database (manual or import)
├── Set up salesman accounts and route assignments
├── Configure credit limits and initial balances
└── Verify data accuracy

Day 5-6: TRAINING
├── Admin training: Full system walkthrough (2–3 hours)
├── Manager training: Daily operations workflow (2 hours)
├── Salesman training: Mobile app order-taking (1 hour per batch)
└── Practice with sample orders and test data

Day 7: GO LIVE
├── Switch to production use
├── Monitor first day's operations
├── Address any issues in real-time
└── Confirm sync is working for all mobile users

Day 8-30: SUPPORT & OPTIMIZATION
├── Daily check-ins for first week
├── Weekly check-ins for first month
├── Fine-tune credit limits based on real data
├── Review dashboard insights with management
└── Gather feedback for continuous improvement
```

Training Modules

Module	Duration	Audience	Topics
System Administration	3 hours	Admin/IT	Installation, configuration, user management, backup
Daily Operations	2 hours	Managers	Orders, deliveries, stock, dashboard interpretation
Financial Management	2 hours	Finance/Admin	Ledger, payments, aging, collections, statements
Mobile App Mastery	1 hour	Salesmen	Login, order-taking, sync, offline usage
Advanced Reporting	1 hour	Managers/Admin	Custom date ranges, route bills, export, print

20. Contact & Next Steps

Ready to Transform Your Distribution Business?

Schedule a Free Demo

Experience the Distribution Management System with your own data and see the difference in real-time.

Channel	Contact
Email	info@ummahtechinnovations.com
Website	ummahtechinnovations.com
WhatsApp	[Contact Number]
Phone	[Contact Number]

What Happens Next?

1. **Schedule a 30-Minute Demo** — We'll show you the system tailored to your industry
2. **Start a Free 14-Day Trial** — Full access, no credit card required
3. **Get Set Up in 7 Days** — We handle installation, data migration, and training
4. **See Results in 30 Days** — Track your ROI from day one

Appendix A: Feature Summary Checklist

Module	Features	Status
Product Management	CRUD, bulk import, multi-price, categories, brands	✅ Complete
Order Management	Multi-channel, lifecycle, discounts, statistics	✅ Complete
Warehouse Management	Multi-warehouse, stock levels, movements, alerts	✅ Complete
Delivery Management	Challans, load sheets, partial delivery, returns	✅ Complete

Module	Features	Status
Financial Ledger	Double-entry, FIFO, aging, credit limits	✅ Complete
Daily Collections	Collection tracking, summaries, reports	✅ Complete
Route Management	CRUD, consolidated bills, statistics	✅ Complete
Shop Management	Profiles, credit limits, contact integration	✅ Complete
Salesman Management	Profiles, targets, performance, credentials	✅ Complete
Mobile App (Offline)	Order-taking, auto-sync, background sync	✅ Complete
Dashboard & Analytics	KPIs, trends, top performers, alerts	✅ Complete
PDF Generation	Thermal receipts, delivery challans	✅ Complete
Role-Based Access	Admin, Manager, Salesman, Warehouse	✅ Complete
Company Settings	Branding, tax, bank, currency configuration	✅ Complete
Stock Returns	Return processing, reason tracking, stock restoration	✅ Complete
Salesman Ledger	Salary, commission, advance, bonus tracking	✅ Complete
GPS Tracking	Geofencing, visit verification	🔄 Roadmap
Barcode Scanning	Product scan for orders	🔄 Roadmap
Push Notifications	Real-time alerts	🔄 Roadmap
iOS App	Apple platform support	🔄 Roadmap
AI Forecasting	Demand prediction, route optimization	🔄 Roadmap
Multi-Language	Urdu, Arabic, Hindi, and more	🔄 Roadmap

Appendix B: Glossary of Terms

Term	Definition
DMS	Distribution Management System — the complete software platform
FMCG	Fast-Moving Consumer Goods — products sold quickly at low cost
Challan	Delivery document listing items being dispatched
Load Sheet	Warehouse loading document consolidating multiple deliveries
Aging Report	Analysis of receivables by days outstanding
FIFO	First In, First Out — payment allocation to oldest invoices first
SKU	Stock Keeping Unit — unique identifier for each product variant
RBAC	Role-Based Access Control — permission system based on user roles
WAL	Write-Ahead Logging — SQLite mode for concurrent read/write

Term	Definition
Thermal Receipt	80mm wide receipt printed on thermal POS printers
Soft Delete	Marking records as inactive instead of permanent deletion
LRU Cache	Least Recently Used Cache — performance optimization technique

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